

Complete the example for each of the following enthalpy changes

Symbol	Definition & example
ΔH_r	Standard enthalpy change of reaction ($\Delta H_r^\ominus$) is the <u>energy change</u> in a chemical reaction when the molar quantities of reactants stated in the chemical equation react at 298 K and 1 bar. $2\text{Na(s)} + 2\text{HCl(aq)} \longrightarrow 2\text{NaCl(aq)} + \text{H}_2\text{(g)}$
ΔH_f	Standard enthalpy change of formation ($\Delta H_f^\ominus$) of a substance is the <u>energy change</u> when one mole of the pure substance in a specified state is formed from its constituent elements in their standard states at 298 K and 1 bar. $\longrightarrow \text{C}_4\text{H}_{10}\text{(g)}$
ΔH_c	Standard enthalpy change of combustion ($\Delta H_c^\ominus$) of a substance is the <u>energy released</u> when one mole of the substance is completely burnt in excess oxygen at 298 K and 1 bar. $\text{C}_4\text{H}_{10}\text{(g)} + \text{.....} \longrightarrow \text{.....}$
ΔH_{neut}	Standard enthalpy change of neutralisation ($\Delta H_{\text{neut}}^\ominus$) is the <u>energy change</u> when an acid and a base react to form one mole of water at 298 K and 1 bar. $\text{NaOH(aq)} + \text{HCl(aq)} \longrightarrow \text{.....}$
ΔH_{atom} (element)	Standard enthalpy change of atomisation of an element ($\Delta H_{\text{atom}}^\ominus$) is the <u>energy absorbed</u> to form one mole of gaseous atoms from the element at 298 K and 1 bar. $\text{Br}_2\text{(l)} \longrightarrow \text{.....}$
ΔH_{atom} (compound)	Standard enthalpy change of atomisation of a compound ($\Delta H_{\text{atom}}^\ominus$) is the <u>energy absorbed</u> to form gaseous atoms from one mole of the compound at 298 K and 1 bar. $\text{C}_4\text{H}_{10}\text{(g)} \longrightarrow \text{.....}$
ΔH_{hyd}	Standard enthalpy change of hydration ($\Delta H_{\text{hyd}}^\ominus$) of an ion is the <u>energy released</u> when one mole of the gaseous ion is dissolved in water to form one mole of aqueous ions at 298 K and 1 bar. $\longrightarrow \text{Na}^+\text{(aq)}$ <i>Note: $\Delta H_{\text{hydration}}$ of an ion $\propto$ charge density</i>

Symbol	Definition & example
ΔH_{soln}	Standard enthalpy change of solution ($\Delta H_{\text{soln}}^\ominus$) of a substance is the <u>energy change</u> when one mole of the substance is completely dissolved in a solvent to form an infinitely dilute solution at 298 K and 1 bar. $\text{NaCl(s)} + \text{aq} \longrightarrow \text{.....}$
BE	Bond energy (BE) of a bond is the <u>average energy absorbed</u> when 1 mole of the bonds are broken in the gaseous state. -
BDE	Bond Dissociation Energy of a bond is the <u>energy required</u> to break 1 mole of that particular bond in a <u>particular</u> compound in gaseous state. $\text{Br}_2 \text{.....} \longrightarrow \text{.....}$
IE	First ionisation energy (1st I.E.) of an element is the <u>energy required</u> to remove one mole of electrons from one mole of gaseous atoms to form one mole of singly charged positive gaseous ions. $\text{Br} \text{.....} \longrightarrow \text{.....}$
EA	First electron affinity (1st E.A.) of an element is the <u>energy change</u> when one mole of gaseous atoms acquires one mole of electrons to give one mole of singly charged negative gaseous ions. $\text{Br} \text{.....} + \text{.....} \longrightarrow \text{.....}$
LE	Lattice energy (LE) of an ionic compound is the <u>energy released</u> when one mole of the solid ionic compound is formed from its constituent gaseous ions at 298 K and 1 bar. $\longrightarrow \text{NaCl(s)}$

Calculating ΔH from experiment

10 cm³ of 1 mol dm⁻³ NaOH was added to 20 cm³ of 0.2 mol dm⁻³ of H₂SO₄.

The temperature of the solution rose from an initial 30.0 °C to 33.2 °C.

➤ Calculate the $\Delta H_{\text{neutralisation}}$

Assume that the specific heat capacity and density of all solutions are

4.18 J g⁻¹ K⁻¹ and 1.00 g cm⁻³ respectively

Step 1: Calculate q

$$q = mc\Delta T$$

Notes

- m = total mass of the solution
Do not include mass of solids
- ΔT = change in temperature
Do not add 273

Step 2: Calculate ΔH_{neut}

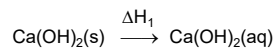
$$\Delta H = -\frac{q}{n_A} \times a$$

Notes

- n_A = amount of A (in mol)
- a = stoichiometric coefficient of A
 - for ΔH_{neut} : $A = \underline{\hspace{2cm}}$
 $a = \underline{\hspace{2cm}}$
 - for ΔH_c : A = substance combusted
 $a = \underline{\hspace{2cm}}$
 - for ΔH_f : $A = \underline{\hspace{2cm}}$

Energy cycle

Using the data in the table below, draw an energy cycle to calculate ΔH_1 .



enthalpy	$\Delta H / \text{kJ mol}^{-1}$	equation
ΔH_f of H ₂ O(l)	-286	
ΔH_f of Ca(OH) ₂ (s)	-987	
ΔH_f	-408	$\text{Ca(s)} + 2\text{H}_2\text{O(l)} \longrightarrow \text{Ca(OH)}_2(\text{aq}) + \text{H}_2(\text{g})$

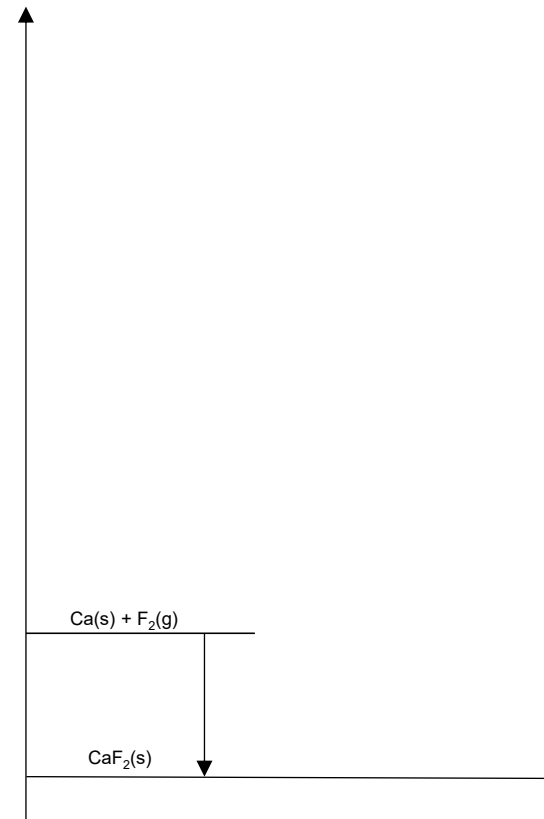
Checklist

- all equations balanced
- state symbols for all species
- all arrows labelled with specified ΔH or value
- application of Hess' Law

Born-Haber cycle

Born-Haber cycle refers to energy level diagram

Complete the Born-Haber cycle for CaF₂(s)



Checklist

- all equations balanced with state symbols
- labelled vertical axis
- labelled 0 kJ mol⁻¹
- all arrows labelled with specified ΔH or value
- arrows point $\uparrow$ for +ve ΔH
- arrows point $\downarrow$ for -ve ΔH
- arrow head and tail touches the energy levels
- longer arrows for larger magnitude

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ΔH_f	Standard enthalpy change of formation ($\Delta H_f^\ominus$) of a substance is the <u>energy change</u> when one mole of the pure substance in a specified state is formed from its constituent elements in their standard states at 298 K and 1 bar. $\dots\dots\dots 4\text{C(s)} + 5\text{H}_2\text{(g)} \longrightarrow \text{C}_4\text{H}_{10}\text{(g)}$
ΔH_c	Standard enthalpy change of combustion ($\Delta H_c^\ominus$) of a substance is the <u>energy released</u> when one mole of the substance is completely burnt in excess oxygen at 298 K and 1 bar. $\text{C}_4\text{H}_{10}\text{(g)} + \dots\dots\dots \frac{13}{2}\text{O}_2\text{(g)} \longrightarrow \dots\dots\dots 4\text{CO}_2\text{(g)} + 5\text{H}_2\text{O(l)}$
ΔH_{neut}	Standard enthalpy change of neutralisation ($\Delta H_{\text{neut}}^\ominus$) is the <u>energy change</u> when an acid and a base react to form one mole of water at 298 K and 1 bar. $\text{NaOH(aq)} + \text{HCl(aq)} \longrightarrow \dots\dots\dots \text{NaCl(aq)} + \text{H}_2\text{O(l)}$
ΔH_{atom} (element)	Standard enthalpy change of atomisation of an element ($\Delta H_{\text{atom}}^\ominus$) is the <u>energy absorbed</u> to form one mole of gaseous atoms from the element at 298 K and 1 bar. $\dots\dots\dots \frac{1}{2}\text{Br}_2\text{(l)} \longrightarrow \dots\dots\dots \text{Br(g)}$
ΔH_{atom} (compound)	Standard enthalpy change of atomisation of a compound ($\Delta H_{\text{atom}}^\ominus$) is the <u>energy absorbed</u> to form gaseous atoms from one mole of the compound at 298 K and 1 bar. $\text{C}_4\text{H}_{10}\text{(g)} \longrightarrow \dots\dots\dots 4\text{C(g)} + 10\text{H(g)}$
ΔH_{hyd}	Standard enthalpy change of hydration ($\Delta H_{\text{hyd}}^\ominus$) of an ion is the <u>energy released</u> when one mole of the gaseous ion is dissolved in water to form one mole of aqueous ions at 298 K and 1 bar. $\dots\dots\dots \text{Na}^+\text{(g)} \longrightarrow \text{Na}^+\text{(aq)}$ <i>Note: $\Delta H_{\text{hydration}}$ of an ion $\propto$ charge density</i>

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LE	Lattice energy (LE) of an ionic compound is the <u>energy released</u> when one mole of the solid ionic compound is formed from its constituent gaseous ions at 298 K and 1 bar. $\dots\dots\dots \text{Na}^+\text{(g)} + \text{Cl}^-\text{(g)} \longrightarrow \text{NaCl(s)}$

